1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive urothelial carcinoma (MIBC), tumors with high intratumoral T cell infiltration but poor response to PD-1 blockade (pembrolizumab) harbor elevated LAG-3 pathway activity that maintains T cell exhaustion despite PD-1 inhibition. Consequently, high TIL density combined with high LAG-3 expression predicts reduced pembrolizumab efficacy, independent of PD-L1 status. Modulation of the local immune milieu by intravesical BCG can alter the balance of exhaustion and activation such that PD-1 blockade alone remains insufficient in LAG-3–high contexts, whereas dual PD-1/LAG-3 blockade or sequential therapy incorporating BCG improves anti-tumor responses. In contrast, LAG-3–low tumors among TIL-rich MIBC patients are more likely to benefit from PD-1 blockade alone. This framework supports a biomarker-driven clinical strategy that stratifies patients by TIL density and LAG-3 expression to guide monotherapy versus combination therapy (PD-1/LAG-3) and sequencing with BCG.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	LAG3
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

LAG3: LAG3 can be expressed by tumor or tumor-infiltrating cells support a context-dependent inhibitory role; the BCG relationship provides a plausible modulator of the local immune environment.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	BCG	Supporting modifier from knowledge graph
Secondary	urothelial carcinoma (BLCA)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- BCG: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- urothelial carcinoma (BLCA): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.